

Plug Puller — Starter Guide

From “that plug is stuck” to a printed tool: figure out your plug (match it or measure it), fill in the Customizer steps, print.

The last two pages of this PDF are 1:1 measuring stencil sheets. Print this document at 100% scale / “Actual size” — never “fit to page” — and check the 50 × 50 mm calibration square on each of those pages before trusting it.

What you need

The plug you want to pull (in its outlet), and either the printed **measuring stencil** (stl/Measuring-Stencil/, 1.2 mm cards, no supports) or the paper stencil sheet at the back of this document.

The stencil cards — legend

Each 3D-printed card has a raised two-letter ID you can find by touch:

P1	plug silhouette: flat 2-prong lamp plug (NEMA 1-15)
P2	plug silhouette: standard 3-prong plug (NEMA 5-15)
P3	plug silhouette: heavy-duty extension cord (NEMA 5-15)
P4	plug silhouette: wide 2-prong appliance plug (NEMA 1-15)
R1	ruler — raised mm ticks, numerals every 10 mm, tactile edge notches
C1	cord gauge — open slots Ø 3–9 mm; slide it onto the cord from the side
F1 / F2	finger sizing — 18 labeled holes (Ø 15–25 on F1, Ø 26–32 on F2)

Each P card has three openings: **W** (the plug's width outline), **T** (its thickness outline), and an open cord slot. It is a “does my plug match this preset?” test — no numbers involved.

Path A — match a preset (fastest, no numbers)

1. Take the **P1**, **P2**, **P3** and **P4** cards to your plug.
2. Hold each card's **W** cutout over the plug (wide side), then the **T** cutout (thin side); slide the cord slot onto the cord.
3. If the plug fills one card's openings — snug, no big gaps — that preset is your plug. Pick it in Step 1:
 - P1 → Flat 2-prong lamp plug - NEMA 1-15
 - P2 → Standard 3-prong plug - NEMA 5-15
 - P3 → Heavy-duty extension cord - NEMA 5-15 in the two-sided puller
 - P4 → Wide 2-prong appliance plug - NEMA 1-15
4. Skip ahead to “Fill in the Customizer steps” on the next page.

No card fits? Your plug is between presets — take Path B; the measured tool will fit better than any preset anyway.

Path B — measure (about 5 minutes)

Work the worksheet in docs/guides/measuring-guide.md; the numbers below are the worksheet's own numbering.

1	Plug length — R1 , wall plate to the plug's back face
2–3	Plug width at the prong end / at the cord end — R1 notched edge against the plug body
4–5	Plug thickness at the prong end / at the cord end — R1 , across the thin direction
6	Cord thickness — C1 , slide it onto the cord from the side; the smallest slot that slips over
7	Wall plate style — a picture quiz, see the measuring guide
8	Finger knuckle width — F1 / F2 , smallest comfortable hole, minus 5
9	Hand width — R1 , across the four knuckles, flat hand

8 and 9 are only needed if you pick “Measure my hand” in Step 2 — the built-in Small / Medium / Large cover most hands.

Fill in the Customizer steps

The Plug Puller comes as two tools, each in its own file. The **one-sided puller** (`src/Plug_Puller_Parametric.scad`) fits lamp plugs (P1), standard 3-prong plugs (P2), the wide 2-prong appliance plug (P4) and other plugs thinner than 24 mm. The **two-sided puller** (`src/Plug_Puller_Two_Sided.scad`) closes two plates on the plug from both sides: thick round plugs like P3, the USB-C laptop tip, charger plugs, and the lamp and standard plugs too (both files offer those two as presets). A plug 24 mm thick or more gets a red tag in the one-sided file that sends you to the two-sided file.

Open your tool's file in OpenSCAD and show the Customizer panel (uncheck *View ▸ Hide Customizer*). Every click is spelled out in `docs/guides/quick-start-beginner.md`; the short version:

Step	One-sided puller	Two-sided puller
1 — Your Plug	the preset from Path A, or your Path B numbers with <code>plug_preset = Measure my plug</code>	a preset: Heavy-duty extension cord - NEMA 5-15 for a P3 plug, USB-C laptop tip, or the lamp or standard plug; or <code>Measure my plug</code> and four numbers: length, width at the prong end and at the cord end (the size the two plates close across), cord. Then <code>Rounded sides</code> (a round plug, a USB-C or charger tip: a sloped cradle centers it) or <code>Flat sides</code> (a boxy plug: the teeth bite along the whole side)
2 — Size	Medium fits most adults; or <code>Measure my hand</code> with worksheet numbers 8–9	the same; <code>Measure my hand</code> asks for number 8 only
3 — Attachment	keep the default <code>Zip ties + Velcro</code> unless you know you want less	<code>Zip ties + Velcro strap</code> (default) or <code>Zip ties</code> : the zip ties hold the two plates together
4	Cord Hook: Right or Left	Print Layout: keep Both plates, one file prints the whole tool

Press **F6** to render, then *File ▸ Export ▸ STL*. For the two-sided puller: print both plates, flip one over, zip-tie the pair face to face around the plug.

Which steps shape which tool

Step	One-sided puller	Two-sided puller
1 — Your plug	pocket, notch, hook slot	arm gap, arm length, cord channel, cradle
2 — Size	finger holes, body size	finger holes
3 — Attachment	zip-hole grid, velcro wings	zip stations, arm strap slots
4	Cord hook: J-hook direction	Print layout: both plates or one

Printing the 3D stencil on a small bed

The cards pack onto print sheets automatically. Open `Measuring_Stencil.scad`, set `bed_width` and `bed_depth` to your printer's bed, and the console lists which cards land on which sheet. Render with `part_index = 1`, export and print, then repeat with `part_index = 2`, and so on. (`part_index = 0` previews all sheets at once — don't print that one.)

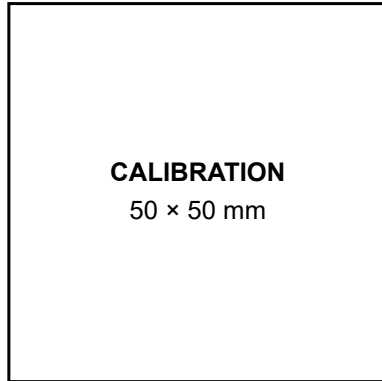
If the print doesn't fit

One symptom, one number to nudge: `docs/guides/fit-troubleshooting.md`. Want to sanity-check before printing the real tool? Print a paper 1:1 outline sheet of your combination first (`docs/guides/print-preview-outlines.md`).

Next pages: the 1:1 measuring stencil sheets (the wide 2-prong appliance plug, P4, is on the second one). Cut along the dashed lines. Measure the calibration square on each page before trusting anything — it must be exactly 50 × 50 mm.

Plug Puller — Measuring Stencil Sheet

Print at 100% scale / “Actual size” — never “fit to page”. Check the calibration square before trusting anything on this sheet.

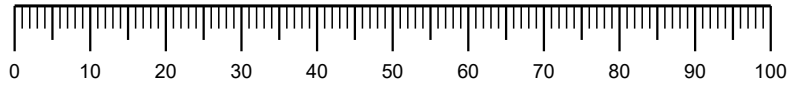


1. Measure this square with a ruler.
It must be exactly 50 × 50 mm — if not, your print was scaled: re-print at 100%.

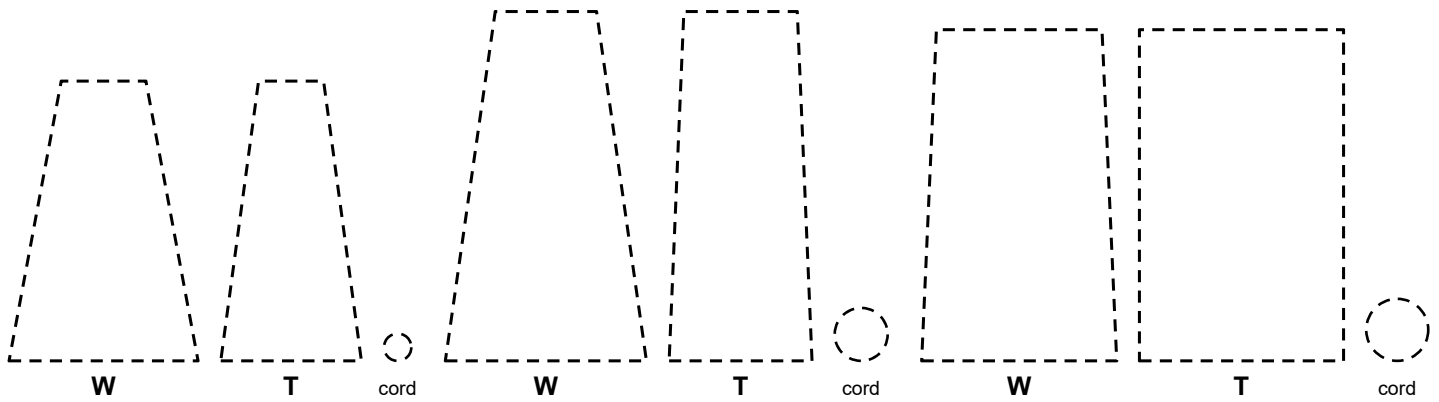
Card legend: <docs/guides/starter-guide.md>

Works on A4 and US Letter paper.

R1 — ruler (mm): plug length, widths, thicknesses, hand width



2. Plug silhouettes (1:1) — hold your plug against **W** (wide side) and **T** (thin side); if it matches, pick that preset in Step 1.

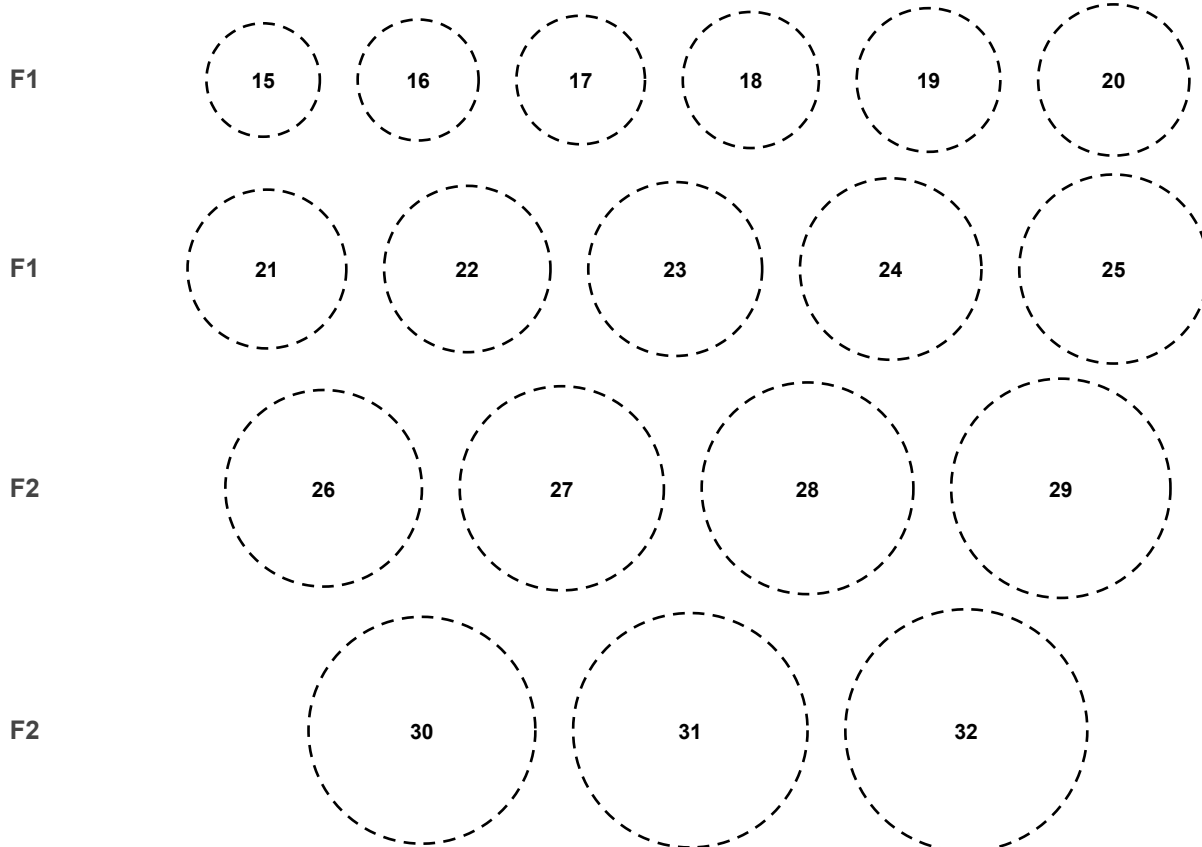


P1 Lamp 2-prong NEMA 1-15

P2 Standard 3-prong NEMA 5-15

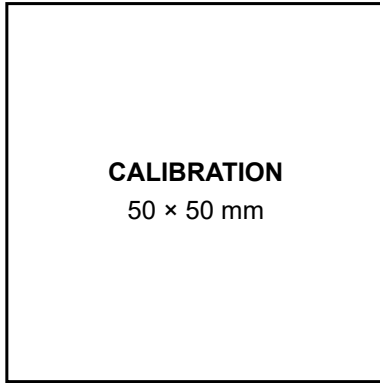
P3 Heavy-duty cord NEMA 5-15

3. **F1 / F2 — finger circles: smallest circle your middle finger passes through, minus 5 = finger width.**



Plug Puller — Measuring Stencil Sheet

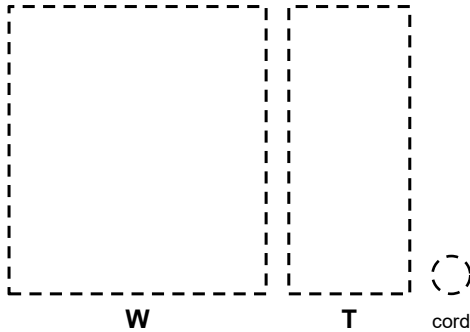
Print at 100% scale / “Actual size” — never “fit to page”. Check the calibration square before trusting anything on this sheet.



1. Measure this square with a ruler.
It must be exactly 50 × 50 mm — if not, your
print was scaled: re-print at 100%.

Card legend: docs/guides/starter-guide.md
Works on A4 and US Letter paper.

2. Plug silhouettes (1:1), continued — hold your plug against W (wide side) and T (thin side); if it matches, pick that preset in Step 1.



P4 Wide 2-prong NEMA 1-15